MIL-M-60145A(MU)
11 July 1966
SUPERSEDING
MIL-M-60145(MU)
7 January 1965

MILITARY SPECIFICATION

MOTOR, CONTROL: 10527239
(PITCH TORQUER)

1. SCOPE

1.1 This specification covers one type of control motor that supplies torque to the elevation gimbal of a gyro-stabilized sight assembly employed in a UH-1B Helicopter.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

C10527239	Motor, Control: 10527239 (Pitch Torquer)

FSC 1270

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PACKAGING DATA SHEET

MIL-P-14232/P10527239 Packaging of Motor, Control 10527239

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Design and construction.- The control motor shall be designed and constructed in accordance with Drawing C10527239 and the requirements of this specification.

3.2.1 Treatment, moisture and fungus-resistant.- The control motor shall be treated for moisture and fungus resistance in accordance with MIL-T-152.

3.2.2 Inductance.- The inductance of the motor armature between brush terminals shall be 0.012 henries (H) ± 30% at a temperature of 80° ± 10° Fahrenheit (F).

3.2.3 Resistance.- The direct current (dc) resistance between the brush terminals shall be 13.0 ohms ± 12.5% at 80° ± 10°F.

3.2.4 Insulation resistance.- The resistance between any one terminal and the motor housing shall not be less than 100 megohms when 500 Vdc is applied at 150°F.

3.2.5 Direction of rotation.- The motor armature, when viewed from the bell housing end, shall rotate clockwise when a d-c voltage is applied to the number 1 (positive) and number 2 (negative) motor terminals.

3.2.6 Dielectric breakdown.- No insulation breakdown shall occur at a temperature of 150°F when 500 Vac is applied between any one input terminal and the housing.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Part identification and marking
(d) Workmanship
(e) Soldering electrical connections

3.4 Environmental.

3.4.1 Storage temperature.- The control motor shall meet the per-

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formance requirements of 3.5 at 60° to 90° Fahrenheit (F) after being subjected to ambient temperatures at -80° and at +160°F.

3.4.2 Operating temperature.- The control motor shall meet the performance requirements of 3.5 while exposed to temperatures at -40° and at +150° Fahrenheit (F).

3.4.3 Altitude.- The control motor shall meet the performance requirements of 3.5 at an altitude of 30,000 feet above sea level.

3.4.4 Shock.- The control motor shall withstand impact shocks up to 15 G's, each shock impulse having a time duration of 11 ± 1 milliseconds. Subsequently, the control motor shall meet the performance requirements of 3.5.

3.4.5 Vibration.- The control motor shall meet the performance requirements of 3.5 after being subjected to vibrations up to 5 G's over a frequency range of 5 to 500 cycles per second.

3.5 Performance.

3.5.1 Torque sensitivity.- The control motor shall develop a torque of 38.4 ounce-inches (oz-in) +15% per ampere of driving current.

3.5.2 Torque linearity.- The values of torque on the torque-current characteristic line shall not deviate more than 5% from the most linear portion between 0 and 50 oz-in.

3.5.3 Torque variation (ripple).- The torque sensitivity (oz-in per ampere) shall be within 6% of the nominal value over one revolution of the armature.

3.5.4 Friction torque.- The break-away or minimum torque under which armature motion occurs shall be less than 1.65 oz-in. maximum.

3.5.5 Peak torque volts-amperes.- The nominal voltage and rated amperes required to generate full motor torque at 150°F shall be 20.3 volts and 1.6 amperes, respectively, at a duty cycle of 33%.

3.6 Radio interference suppression.- Interference control requirements of the equipment shall conform to MIL-I-6181.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 First article tests.- Unless otherwise specified, the requirements and tests of Table I shall be conducted on first article tests only. In the event that the processes, materials or other changes in design or manufacture occur subsequent to first article approval, the first items so produced shall be submitted to the Government in accordance with 4.2 for approval prior to resumption of production.

4.

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TABLE I

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Altitude	3.4.3	4.6.2.3
302.	Shock	3.4.4	4.6.2.4
303.	Vibration	3.4.5	4.6.2.5
304.	Radio interference suppression	3.6	4.6.4

4.3.2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 1.5 Percent Defective		
101.	Treatment, moisture and fungus resistant	3.2.1	4.6.1
102.	Inductance	3.2.2	4.6.1.1
103.	Resistance	3.2.3	4.6.1.2
104.	Insulation resistance	3.2.4	4.6.1.3
105.	Direction of rotation	3.2.5	4.6.1.4
106.	Dielectric breakdown	3.2.6	4.6.1.5
107.	Torque sensitivity	3.5.1	4.6.3.1
108.	Torque linearity	3.5.2	4.6.3.1
109.	Torque variation (Ripple)	3.5.3	4.6.3.2
110.	Friction torque	3.5.4	4.6.3.3
111.	Peak torque volts-amperes	3.5.5	4.6.3.4
MINOR:	NONE		

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- Three motors shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

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TABLE III

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401. Design and construction	3.2	Applicable drawing - visual
402. General specification	3.3	MIL-F-13926 - visual
403. Temperature (storage)	3.4.1	4.6.2.1
404. Temperature (operating)	3.4.2	4.6.2.2

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Treatment, moisture and fungus-resistant.- The control motor shall be tested in accordance with MIL-T-152 to insure compliance with 3.2.1.

4.6.1.1 Inductance test.- With the control motor stabilized at 80 $\pm$ 10°F, connect an impedance bridge across the brush terminals and measure

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the inductance for compliance with 3.3.2.

4.6.1.2 Resistance test.- With the control motor stabilized as in 4.6.1.2, connect a 1% accuracy ohmmeter across the brush terminals and measure the resistance for compliance with 3.3.3.

4.6.1.3 Insulation test.- With the control motor stabilized at 150°F, connect a megohm bridge between one terminal and the rotor housing and measure the resistance for compliance with 3.3.4.

4.6.1.4 Direction of rotation test.- Apply an input voltage to the control motor with the positive polarity connected to the number 1 designated motor terminal and the return to number 2. Observe the direction of rotation while viewing from the bell housing end for compliance with 3.3.5.

4.6.1.5 Dielectric breakdown.- With the control motor stabilized at 150°F, connect a hi-pot between one input terminal and the housing. Adjust hi-pot for a potential of 500 Vac and check dielectric breakdown for compliance with 3.2.6.

4.6.2 Environmental.

4.6.2.1 Storage temperature.- The control motor, while in a non-operating condition, shall be subjected to the Temperature Cycling Test, Procedure II of MIL-F-13926 using Conditions A and H. Upon stabilization at ambient room temperature, the control motor shall meet the performance requirements specified in 3.5.

4.6.2.2 Operating temperature.- The control motor shall be subjected to the Temperature Cycling Test, Procedure I of MIL-F-13926 using Conditions C and G. During temperature stabilization at Conditions C, G and room ambient, the control motor shall meet the performance requirements specified in 3.5.

4.6.2.3 Altitude test.- The control motor shall be placed within a test chamber and the absolute internal pressure reduced to 8.88 inches of mercury (corresponding to an altitude of 30,000 feet above sea level). After sufficient time has elapsed to allow all entrapped air to escape, the control motor shall meet the requirements specified in 3.5.

4.6.2.4 Shock test.- The control motor shall be tested in accordance with MIL-F-13926, Procedure I. The control motor shall not suffer damage or subsequently fail to provide the performance specified in 3.5.

4.6.2.5 Vibration test.- With the control motor mounted and operating at room ambient conditions, the table shall be vibrated along a horizontal direction with the frequency varying between 5 and 500 cps. For frequencies between 5 and 20 cps, the double amplitude shall be 0.1 inch. For frequencies between 20 and 33 cps, the amplitude shall be varied so as to maintain a 2G acceleration. For frequencies between 33 and 52 cps,

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the double amplitude shall be 0.036 inch. For frequencies between 52 and 500 cps, the amplitude shall be varied so as to maintain a 5G acceleration. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (5-500-5) will consume approximately 15 minutes. The test shall continue for four 15-minute cycles for a total of 1 hour. (When provisions for logarithmic cycling is not available, a linear rate of frequency change may be used.) The cycling test shall be repeated with the axis of vibration changed 90 degrees in the horizontal direction and then with the axis of vibration changed to vertical. At the completion of each direction of vibration, the control motor shall be closely inspected for any evidence of mechanical failure, loose or broken parts. At the end of test, the control motor shall meet the requirements specified in 3.5.

4.6.3 Performance.

4.6.3.1 Torque sensitivity and linearity test.- Mount the control motor in a test device as depicted in Figure 1. Take the current readings versus the load points for 0, 5, 10, 20, 40, 60 and 80 ounces. Reverse the current source and repeat the test. Make a graphical plot of torque (ounces times torque arm in inches) versus average current. Establish the most favorable line of the plotted points. The resultant plot shall meet the requirements specified in 3.5.1 and 3.5.2.

4.6.3.2 Torque variation test.- With the control motor connected as in 4.6.3.1, energize the control motor with a current of 1.0 ampere. Measure the spring deflection for four-compass points of the armature which are a known equal distance from the center of rotation. Repeat for both clockwise and counter-clockwise rotations and average the resultant readings. Plot the points and establish the best straight line. The resultant plot shall meet the requirement specified in 3.5.3.

4.6.3.3 Friction torque test.- With the control motor connected as in 4.6.3.1, the torquer sensitivity having been established, measure the breakaway or minimum current at which motion of the armature occurs. Repeat three times for both clockwise and counter-clockwise rotation and average the readings. Then using the established torquer sensitivity of 4.6.3.1, calculate the friction torque to insure compliance with 3.5.4.

4.6.3.4 Peak torque volts-amperes test.- Connect the control motor as in 4.6.3.1 with the addition of a 1.0 percent accuracy voltmeter connected across the control motor input. Install in a hot chamber stabilized at 150°F. Apply 1.6 amperes to the control motor and observe the scale deflection for maximum torque and simultaneously observe the voltage reading. Verify compliance with 3.5.5.

4.6.4 Radio interference suppression test.- The equipment shall be tested in accordance with MIL-I-6181.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10527239. The level of protection shall be as specified in the procurement document.

8.

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6. NOTES

6.1 Intended use.- The Control Motor: 10527239 is part of the Sight Assembly: 10527236 which is part of the Antioscillation Sighting System XM58 employed in a UH-1B Helicopter. The torque supplied by the control motor is intended to counteract any disturbance torques introduced as a result of bearing friction caused by aircraft motion or other unbalancing torques.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing (see 5.1).
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1270-A034Y

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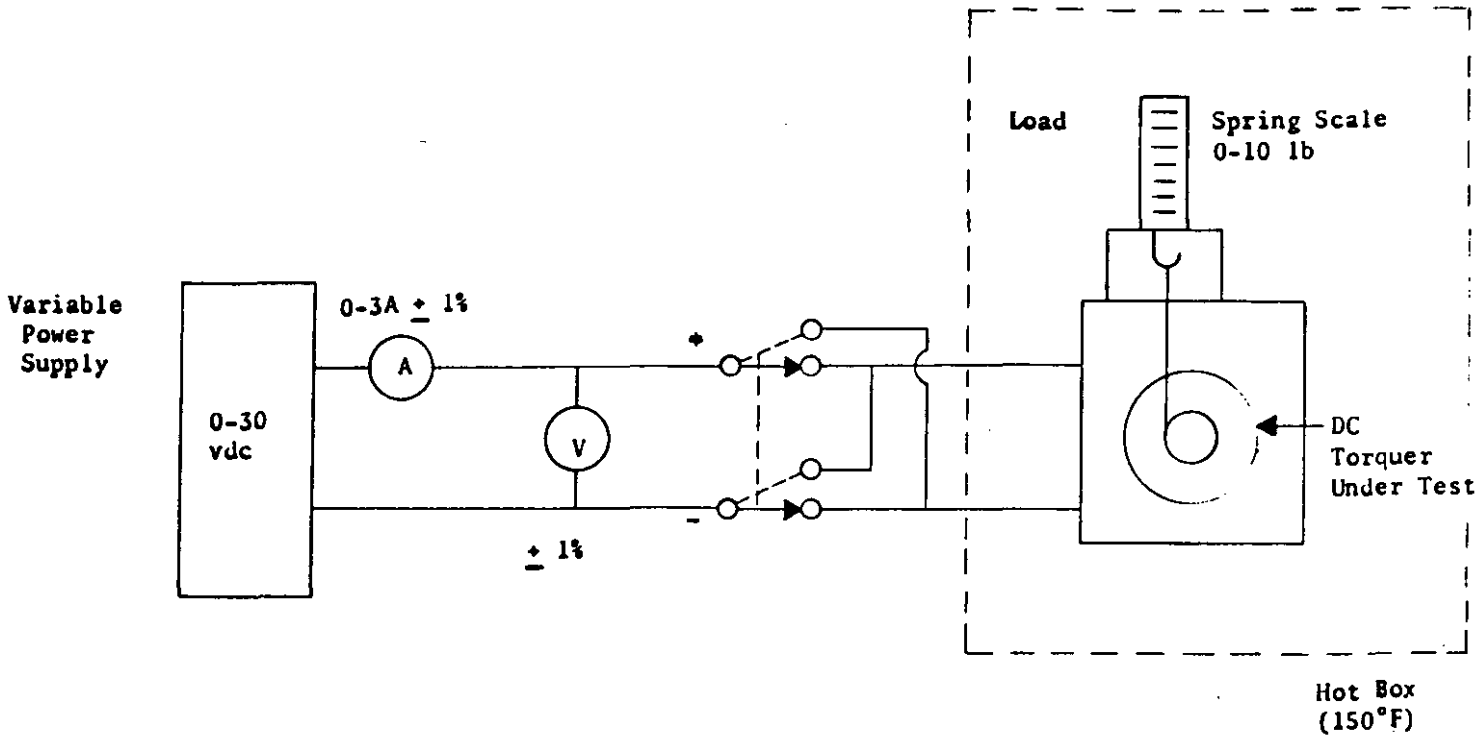


FIGURE 1 - Test Setup for Control Motor

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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